Name _____

Quiz 4—Alternate

Fill in the missing parts of the proof by induction of the theorem below.

Theorem: For every tree (connected, undirected graph with no cycles), the number of edges is one less than the number of vertices.

Proof: (2 points) Universal declaration: Let T be an arbitrary

We induct on n, the number of nodes in T.

(3 points) Induction hypothesis:

There are two cases:

- Base case: Suppose $n = 1$. Then T is a single node with no edges, so the number of edges is 0, which is one less than the number of vertices, which is 1.
- (5 points) Inductive case: Suppose

The two cases are exhaustive, and the claim holds in both cases. Therefore, the claim hods for all $n \geq 1$.

□